Prisma

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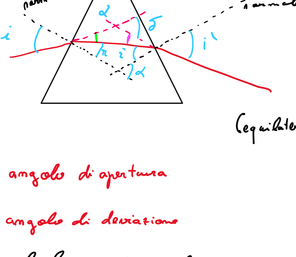


(equilatero in teoria)

α: angolo di apertura

δ: angolo di deviazione

Per le leggi di snell:

$\operatorname{sen} i = n \operatorname{sen} r \qquad \operatorname{sen} i' = n \operatorname{sen} r'$

$r + r' = \alpha \qquad \delta = (i - r) + (i' - r') = i + i' - \alpha$

Deviazione minima ra

$$\frac{d\delta}{di} = 0 \Rightarrow \frac{d\delta}{di} = 1 + \frac{di'}{di} = 0 = \boxed{\frac{di'}{di} = -1}$$

Differenziamo le relazioni di snell:

$$\begin{cases} \cos i \, di = n \cos r \, dr \quad e \quad \cos i' \, di' = n \cos r' \, dr' \\ r + r' = \alpha \Rightarrow dr + dr' = 0 \Rightarrow dr = -dr' \end{cases}$$

$$di = \frac{n \cos r \, dr}{\cos i} \qquad di' = \frac{n \cos r' \, dr'}{\cos i'}$$

$$\Rightarrow \frac{di'}{di} = \frac{-\cancel{n} \cos r' \, \cancel{dr}}{\cos i'} \cdot \frac{\cos i}{\cancel{n} \cos r \, \cancel{dr}} = -1$$

$$\Rightarrow \frac{\cos r' \cos i}{\cos i' \cos r} = 1$$

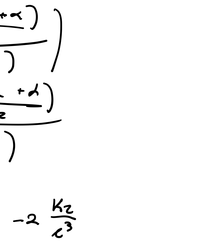
$$\Rightarrow \cos r' \cos i = \cos r \cos i'$$

$$\Leftrightarrow \boxed{i = i'} \quad e \quad \boxed{r = r'}$$

DEVIAZIONE MINIMA:

$i = i'$

$r = r'$



$\Rightarrow 2r = \alpha \qquad r = \frac{\alpha}{2}$

$\delta = 2i - \alpha$

$2i = \delta_{min} + \alpha \qquad i = \frac{\delta_{min} + \alpha}{2}$

$\operatorname{sen} i = n \operatorname{sen} r$ Snell:

$$n = \frac{\operatorname{sen} i}{\operatorname{sen} r} = \frac{\operatorname{sen}\left(\frac{\delta_{min} + \alpha}{2}\right)}{\operatorname{sen} \alpha/2}$$

$\Rightarrow$ Determinando δ_{min} si determina sperimentalmente n

Trasandando assorbimento

$$n^2 = 1 + \frac{K}{\omega_0^2 - \omega^2}$$

$$\Rightarrow n = K_1 + \frac{K_2}{\lambda^2} \quad \text{FORMULA DI CAUCHY}$$

DISPERSIONE

$$D = \frac{d\delta}{d\lambda} = \frac{d\delta}{dn} \cdot \frac{dn}{d\lambda}$$

Se la deviazione è minima

$$\frac{dn}{d\delta} = \frac{d}{d\delta}\left(\frac{\operatorname{sen}\left(\frac{\delta_{min} + \alpha}{2}\right)}{\operatorname{sen}(\alpha/2)}\right)$$

$$= \frac{1}{2} \frac{\cos\left(\frac{\delta_{min} + \alpha}{2}\right)}{\operatorname{sen}(\alpha/2)}$$

$$\frac{dn}{d\lambda} = \frac{d}{d\lambda}\left(K_1 + \frac{K_2}{\lambda^2}\right) = -2\frac{K_2}{\lambda^3}$$

$$\Rightarrow D = \frac{d\delta}{d\lambda} = \frac{1}{\cancel{2}} \frac{\cos\left(\frac{\delta_{min} + \alpha}{2}\right)}{\operatorname{sen}(\alpha/2)} \cdot \left(\frac{-\cancel{2}K^2}{\lambda^3}\right)$$

$$\propto \boxed{\frac{1}{\lambda^3}}$$

$\Rightarrow$ Lunghezze d'onda maggiori sono + disperse